

Machine Learning-Based NIDS Datasets

The datasets on this page are designed for machine learning-based Network Intrusion Detection Systems (NIDS) and are organised into the following high-level collections:

- **NetFlow V3 Datasets:** This collection consists of four datasets in NetFlow format. This is the only version incorporating **temporal features**. These datasets build on the 43 features of V2 by adding **10 temporal NetFlow features**, totaling 53 features. They are provided in CSV format.
- **NetFlow V2 Datasets:** Comprising five datasets in NetFlow format, this collection extends the V1 datasets with 43 enhanced NetFlow features. They are available in CSV format.
- **NetFlow V1 Datasets:** This collection includes five datasets converted from four diverse formats into NetFlow format, featuring 12 basic NetFlow features. They are provided in CSV format.
- **CICFlowMeter Datasets:** Consisting of five datasets, this collection is generated using the CICFlowMeter tool by converting three datasets from various formats into CICFlowMeter format, with 80 features. They are available in CSV format.

For full details and the recommended citation for these datasets, please refer to the corresponding links in the tabs below.

[NetFlow V3 Datasets](#)

[NetFlow V2 Datasets](#)

[NetFlow V1 Datasets](#)

[CICFlowMeter Datasets](#)

NetFlow v3 Datasets

Version 3 of the NetFlow datasets is made up of 53 extended NetFlow features. The details of the datasets are published in:

Majed Luay, Siamak Layeghy, Seyedehfaezeh Hosseininoorbin, Mohanad Sarhan, Nour Moustafa, Marius Portmann, *"Temporal Analysis of NetFlow Datasets for Network Intrusion Detection Systems"*.

(<https://arxiv.org/abs/2503.04404>). Please use the following citation to reference these datasets:

```
@misc{luay2025NetFlowDatasetsV3,  
title = {Temporal Analysis of NetFlow Datasets for Network Intrusion Detection  
Systems},  
author = {Majed Luay and Siamak Layeghy and Seyedehfaezeh Hosseininoorbin and  
Mohanad Sarhan and Nour Moustafa and Marius Portmann},  
year = {2025},  
eprint = {2503.04404},  
archivePrefix= {arXiv},\  
primaryClass = {cs.LG},  
url = {https://arxiv.org/abs/2503.04404}  
}
```

NF-UNSW-NB15-v3 NF-ToN-IoT-v3 NF-BoT-IoT-v3

NF-CSE-CIC-IDS2018-v3

NF-BoT-IoT-v3

Please [click here \(https://rdm.uq.edu.au/files/db0187cc-1f78-4215-8e34-303364498866\)](https://rdm.uq.edu.au/files/db0187cc-1f78-4215-8e34-303364498866) to download the dataset.

An IoT NetFlow-based dataset was generated by expanding the NF-BoT-IoT dataset. The features were extracted from the publicly available pcap files and the flows were labelled with their respective attack categories. The total number of data flows is 16,993,808 out of which 16,881,819 (99.7%) are attack samples and 51,989 (0.3%) are benign. There are four attack categories in the dataset, the table below represents the class distribution of all flows.

Class	Count	Description
Benign	51,989	Normal unmalicious flows
Reconnaissance	1,695,132	A technique for gathering information about a network host and is also known as a probe.
DDoS	7,150,882	Distributed Denial of Service is an attempt similar to DoS but has multiple different distributed sources.

Class	Count	Description
DoS	8,034,190	An attempt to overload a computer system's resources with the aim of preventing access to or availability of its data.
Theft	1,651	A group of attacks that aims to obtain sensitive data such as data theft and keylogging

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